Nawres Alwan

PhD Candidate in Computer Science — Quantum Cryptography, Quantum Machine Learning and Quantum Information Science

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PROFILE

PhD researcher at Universiti Malaya, Malaysia. Doctoral thesis submitted; viva voce examination expected in December 2026. Formal degree completion and certificate issuance pending. Research connects mathematical modelling, quantum algorithms, secure quantum information processing and machine learning, with work on quantum key distribution (QKD), quantum image representation, quantum geometric phases and Hamiltonian dynamics, quantum optimization and variational algorithms, hyperchaotic cryptographic systems, and graph theory with numerical simulation.

Research interests: quantum cryptography and QKD; quantum machine learning and optimization; quantum image representation and processing; quantum geometric methods and simulation; chaos theory, graph theory and numerical methods.

EDUCATION

2021 – Present
PhD in Computer Science (candidate), Universiti Malaya, Malaysia
Faculty of Computer Science and Information Technology. Specialization: Quantum Cryptography and Quantum Computing.
Thesis submitted; viva voce examination expected December 2026. Degree certificate pending.

2012 – 2016
M.Sc., University of Technology, Iraq
Thesis: *Spectral Graph Theory with Applications* — characteristic polynomials and spectral properties of graph classes via adjacency, Laplacian and signless Laplacian matrices; extremal eigenvalues of connected irregular tripartite graphs.

2007 – 2011
B.Sc. in Applied Mathematics, University of Technology, Iraq
Thesis: *Spanning Tree* — methods for computing spanning trees of weighted and unweighted graphs, with a MATLAB program for comparative evaluation.

PUBLICATIONS — PEER-REVIEWED JOURNAL ARTICLES

01 **Multilayered Quantum Computing and Simulation System for Enhanced Image Representation of HSI Based Fourier Transform and Adjacency Matrix.**
N. A. Alwan, S. J. Obaiys, N. M. G. Al-Saidi, N. F. Mohd Noor, and Y. Karaca.
Scientific Reports, 2025. doi:10.1038/s41598-025-24168-4

02 **A Comprehensive Review on Classification of Quantum Image Representations.**
N. A. Alwan, N. M. G. Al-Saidi, S. J. Obaiys, N. F. Mohd Noor, and M. Ahmad.
Archives of Computational Methods in Engineering, 2025. doi:10.1007/s11831-025-10369-7

03 **A Multi-Channel Quantum Image Representation Model with Qubit Sequences for Quantum-Inspired Image and Image Retrieval.**
N. A. Alwan, S. J. Obaiys, N. M. G. Al-Saidi, N. F. Mohd Noor, and Y. Karaca.
AIMS Mathematics, 2025. doi:10.3934/math.2025499

04 **Color Image Encryption Through Multi-S-box Generated by Hyperchaotic System and Mixture of Pixel Bits.**
N. A. Alwan, S. J. Obaiys, N. F. Mohd Noor, N. M. G. Al-Saidi, and Y. Karaca.
Fractals, 2024. doi:10.1142/s0218348x24400395

05 **Coding and Descriptions for the Subset of Twin Star Mean Labeling.**
Asia Pacific Journal of Mathematics, 2024. doi:10.28924/APJM/11-59

06 **A Secure Image Cryptosystem via Multiple Chaotic Maps.**
M. T. Wazi, D. S. Ali, N. M. G. Al-Saidi, and **N. A. Alwan**.
Discrete Mathematics, Algorithms and Applications, Vol. 14, No. 4, 2021. doi:10.1142/S179383092150141X

07 **Tripartite Graphs with Energy Aggregation.**
N. A. Alwan, N. M. G. Al-Saidi, and R. T. Rasheed.
Boletim da Sociedade Paranaense de Matematica, Vol. 38, No. 7, pp. 149-167, 2019.
doi:10.5269/bspm.v38i7.44463

08 **A General Formula for Characteristic Polynomials of Some Special Graphs.**
N. A. Alwan and N. M. Al-Saidi.
Engineering and Technology Journal, Vol. 34, No. 5B, pp. 638-650, 2016.

PUBLICATIONS — CONFERENCE PAPERS AND BOOK CHAPTERS

09 **A Trust Optimization Region of the Quantum Natural Gradient for the Large Damping Domain.**
N. A. Alwan, S. J. Obaiys, N. M. G. Al-Saidi, and R. Chadli.
Computational Science and Its Applications - ICCSA 2026 Workshops.
Springer, Cham, 4 July 2026. doi:10.1007/978-3-032-30497-1_18

10 **A Pseudo Random Number Generator Based on 4D Hyperchaotic Systems, Riddled Basins of Attraction and Advanced Microfluidic Technology.**
N. A. Alwan, S. J. Obaiys, N. M. G. Al-Saidi, N. F. Mohd Noor, and Y. Karaca.
Computational Science and Its Applications - ICCSA 2024 Workshops.
Springer, Cham, 30 July 2024. doi:10.1007/978-3-031-65154-0_6

11 **Secure Image Encryption Using Single-Mode Fiber and Dense Wavelength Division Multiplexing in Chaotic Systems.**
N. M. G. Al-Saidi, S. J. Obaiys, **N. A. Alwan**, A. J. Mohammed, A. K. Farhan, and Y. Karaca.
Springer, 2024. doi:10.1007/978-3-031-65154-0_5

12 **Performance-Enhancing of RSA Public Key via Three-Dimensional Hyperchaotic System.**
N. A. Alwan, A. Y. Yousif, and N. M. Al-Saidi.
AIP Conference Proceedings 2325, 020027, February 2021. doi:10.1063/5.0040397

13 **A New Approach for the Characteristic Polynomial of a Complete Tripartite Graph.**
N. A. Alwan, N. M. Al-Saidi, and W. J. Abdulaa.
AIP Conference Proceedings 2334, 080002, March 2021. doi:10.1063/5.0042627

14 **Image Encryption Based on Highly Sensitive Chaotic System.**
D. S. Ali, **N. A. Alwan**, and N. M. Al-Saidi.
AIP Conference Proceedings 2183, 080007, 2019. doi:10.1063/1.5136200

15 **The Energy of a Tripartite Graph.**
N. A. Alwan, N. M. G. Al-Saidi, and R. T. Rasheed.
AIP Conference Proceedings 2091, 030006, April 2019. doi:10.1063/1.5095091

MANUSCRIPTS UNDER REVIEW (NOT YET PUBLISHED)

P1 **Quantum Encryption Framework via Hyperchaotic Systems and Geometric Phase Randomness for Secure Quantum Images.**
Quantum Information Processing, 2025.

P2 **Geometry-Driven Quantum Randomness: Phase-Sensitivity Amplification and Min-Entropy Certification under Hyperchaotic Control.**
Scientific Reports, 2026.

P3 **A Novel Sector-Conditioned Finite-Key Method for Secure Dynamically Phase-Modulated Quantum Key Distribution Using PSB Geometric Phase.**
Scientific Reports, 2026.

INTELLECTUAL PROPERTY

Patent-related institutional submission, Universiti Malaya — *Relational Quantum Geometry for Image and Spatially Indexed Multidomain Data Processing*. 21 September 2026. Submitted.

Patent-related institutional submission, Universiti Malaya — *Quantum geometric learning system and method for structure-aware image processing based on quantum image representation and Hamiltonian-driven manifold evolution*. 2 September 2026. Submitted.

Copyright institutional submission, Universiti Malaya — *Quantum Deep Learning Strategies for Facial Feature Extraction Based on Quantum Image Representation*. 27 September 2026. Submitted.

FUNDING

Universiti Malaya	Two Graduate Research Assistantship grants in support of doctoral research in quantum cryptography and computational quantum science.

TEACHING EXPERIENCE

Lecturer	**Mathematics, Image Processing and Programming** Applied mathematics; image processing; MATLAB and scientific programming; computational methods. *Prepared to teach:* Python and numerical simulation; quantum computing and algorithms; quantum machine learning; quantum cryptography and QKD.

PROFESSIONAL EXPERIENCE

Jan 2025 – Present	**Graduate Research Assistant**, Universiti Malaya (scheduled end: December 2026)
2024 – Present	**IT Department Manager**, Hezn Al-Aman for Trading and General Contracting
2016 – 2024	**Director**, Solution Center for Research and Software

TECHNICAL SKILLS

Programming	Python, MATLAB, JavaScript, TypeScript, HTML, CSS, Lean 4
Quantum and scientific	Qiskit, numerical simulation, algorithm development, LaTeX, scientific writing
AI and ML	PyTorch, TensorFlow, OpenCV, scikit-learn, Hugging Face, Tesseract OCR
Web	Django, Next.js

CONFERENCES, MEMBERSHIPS AND LANGUAGES

Conferences	ICMS 2018, 2019, 2020 (International Conference of Mathematical Sciences); 3rd Scientific Conference on Computing and Information Technology, Anbar University (2019); ICCSA 2024 and ICCSA 2026 Workshops (Springer).
Memberships	IEEE Member; Iraqi Al-Khwarizmi Association (2018–Present).
Languages	Arabic (native); English (professional working proficiency).